

| Training                             | OOD Score     | CIFAR-10 (ID)    |                  |                  | CIFAR-100 (ID)   |                  |                  |
|--------------------------------------|---------------|------------------|------------------|------------------|------------------|------------------|------------------|
|                                      |               | MNIST            | SVHN             | CIFAR-100        | MNIST            | SVHN             | CIFAR-10         |
| -                                    | Teacher MSP   | 0.92/0.56        | 0.90/0.60        | 0.88/0.62        | 0.70/0.96        | 0.81/0.81        | 0.79/0.79        |
| <b>Linear Student</b>                |               |                  |                  |                  |                  |                  |                  |
| CE                                   | Max Diff      | 0.33/0.99        | 0.46/0.94        | 0.36/0.98        | 0.62/0.95        | 0.72/0.89        | 0.74/0.82        |
|                                      | Abs. Max Diff | 0.34/0.99        | 0.46/0.94        | 0.36/0.98        | 0.38/1.00        | 0.28/0.98        | 0.26/0.99        |
|                                      | KL Div.       | 0.74/0.82        | 0.50/0.97        | 0.59/0.89        | 0.71/0.79        | 0.33/1.00        | 0.33/0.98        |
|                                      | Logit L2      | 0.38/0.99        | 0.24/1.00        | 0.37/0.96        | 0.81/0.94        | 0.17/1.00        | 0.28/0.99        |
| Logit MSE                            | Max Diff      | 0.17/1.00        | 0.40/0.96        | 0.35/0.98        | 0.67/0.95        | <b>0.79/0.82</b> | <b>0.78/0.79</b> |
|                                      | Abs. Max Diff | 0.38/0.99        | 0.48/0.96        | 0.45/0.98        | 0.33/1.00        | 0.21/0.99        | 0.22/1.00        |
|                                      | KL Div.       | <b>0.90/0.30</b> | 0.53/0.95        | 0.57/0.92        | 0.44/0.96        | 0.24/1.00        | 0.25/0.99        |
|                                      | Logit L2      | <b>0.95/0.24</b> | 0.40/0.96        | 0.48/0.93        | 0.51/1.00        | 0.21/1.00        | 0.25/1.00        |
| <b>MLP Student</b>                   |               |                  |                  |                  |                  |                  |                  |
| CE                                   | Max Diff      | 0.40/0.98        | 0.56/0.94        | 0.44/0.96        | <b>0.70/0.77</b> | 0.62/0.91        | 0.69/0.84        |
|                                      | Abs. Max Diff | 0.48/0.98        | 0.58/0.94        | 0.50/0.96        | 0.38/1.00        | 0.38/0.97        | 0.35/0.99        |
|                                      | KL Div.       | 0.82/0.63        | 0.69/0.89        | 0.71/0.83        | <b>0.87/0.55</b> | 0.52/0.99        | 0.50/0.97        |
|                                      | Logit L2      | 0.74/0.72        | 0.33/0.96        | 0.50/0.90        | <b>0.94/0.32</b> | 0.31/0.99        | 0.44/0.97        |
| Logit MSE                            | Max Diff      | 0.31/0.96        | 0.42/0.96        | 0.38/0.96        | 0.61/0.97        | 0.74/0.87        | 0.75/0.83        |
|                                      | Abs. Max Diff | 0.71/0.90        | 0.71/0.93        | 0.69/0.92        | 0.39/1.00        | 0.26/1.00        | 0.25/1.00        |
|                                      | KL Div.       | 0.75/0.83        | 0.72/0.94        | 0.72/0.90        | 0.55/0.95        | 0.32/0.99        | 0.30/0.99        |
|                                      | Logit L2      | 0.63/0.91        | 0.59/0.97        | 0.64/0.94        | 0.58/1.00        | 0.28/1.00        | 0.29/1.00        |
| <b>Layer3 Feature Linear Student</b> |               |                  |                  |                  |                  |                  |                  |
| CE                                   | Max Diff      | 0.83/0.78        | 0.90/0.60        | 0.84/0.67        | 0.41/0.97        | 0.67/0.78        | 0.62/0.87        |
|                                      | Abs. Max Diff | 0.87/0.78        | <b>0.91/0.60</b> | 0.87/0.67        | 0.59/0.99        | 0.36/0.99        | 0.39/0.98        |
|                                      | KL Div.       | 0.88/0.80        | <b>0.92/0.65</b> | <b>0.89/0.63</b> | 0.65/0.98        | 0.47/0.98        | 0.51/0.97        |
|                                      | Logit L2      | 0.37/1.00        | 0.60/0.98        | 0.67/0.94        | 0.27/1.00        | 0.26/1.00        | 0.31/1.00        |
| Logit MSE                            | Max Diff      | 0.82/0.68        | 0.82/0.69        | 0.81/0.67        | 0.55/0.96        | 0.76/0.81        | 0.75/0.81        |
|                                      | Abs. Max Diff | 0.89/0.69        | 0.89/0.70        | 0.88/0.68        | 0.45/0.97        | 0.24/1.00        | 0.25/0.99        |
|                                      | KL Div.       | 0.90/0.74        | 0.90/0.70        | <b>0.89/0.67</b> | 0.56/0.98        | 0.28/0.99        | 0.30/0.99        |
|                                      | Logit L2      | 0.50/1.00        | 0.63/0.97        | 0.73/0.90        | 0.49/1.00        | 0.23/1.00        | 0.25/1.00        |